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Hydrogen-Based Power Plants: Optimal Design Using State-Of-The-Art Technologies to Decarbonize Hard-To-Abate Industries

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Abstract

Hard-to-abate industries, including steel, cement, and datacenters, are responsible for approximately 30% of global CO₂ emissions and present intrinsic challenges for decarbonization due to their reliance on high-temperature heat, continuous processes, and carbon-intensive feedstocks. Achieving the climate goals of the Paris Agreement, requires reducing industrial emissions by up to 90% by 2050, as industrial CO₂ could otherwise increase by 12–20% under a business-as-usual scenario. Hydrogen produced via renewable and nuclear energy sources emerges as a pivotal enabler for this transformation: in fact, as energy vector, it allows to overcome both the slow dynamics of nuclear plants and the intermittent and aleatory nature of renewable power plants while it provides a low-carbon, high-energy-density alternative to fossil fuels in energy-intensive processes. In this work, solutions will be presented to decarbonize hard-to-abate industries based on optimal operation and design aimed at costs minimization.

The hydrogen value chain requires specialized infrastructure to ensure technical efficiency and economic feasibility. In the production phase, electrolyzers powered by renewable or nuclear energy are coupled with reciprocating or centrifugal compressors and asset performance management (APM) solutions to optimize hydrogen handling and system reliability. Transportation and storage involve centrifugal pumps and expanders to regulate flow and pressure during delivery, while sensor solutions enable continuous leak detection and predictive maintenance. At the utilization stage, hydrogen-fueled gas turbines can deliver high-efficiency, low-carbon power and process heat. Throughout the value chain, advanced Energy Management System (EMS) platforms guarantee the minimization of the Levelized Cost of Hydrogen (LCOH). Each component of the value chain is critical, and an integrated optimal approach, one that, using advanced algorithms for operation and design, seamlessly combines all these complex technologies to minimize the costs of the solution, is key to achieve technical viability and economic competitiveness for hydrogen-based infrastructures, enabling scalable decarbonization solutions.

To demonstrate the potential of hydrogen-based technologies, integrated solutions suited for real-world applications in hard-to-abate industries are presented. Specifically, this work shows the optimal design of hydrogen-based infrastructures to decarbonize steel factories, combining electrolyzers, battery energy storage systems (BESS), hydrogen compression, storage technologies, renewable energy sources (RES) and hydrogen-fueled gas turbines. The selected optimal mixes provide the most cost-effective plant for different

decarbonization targets. These examples highlight how a coordinated, optimal deployment of different technologies can enable effective and scalable decarbonization pathways.

The optimal design and operation of hydrogen-based hybrid power plants applied to steel factories represents a new kind of approach, which is paramount to minimize the LCOH. The results underscore the importance of designing hydrogen ecosystems that optimize each phase of the value chain, ensuring sustainability, reliability, and cost-efficiency as we transition to a low-carbon industrial future.

Introduction

The urgency of global decarbonization has intensified in recent years, driven by the escalating impacts of climate change and the need to meet international climate targets such as those outlined in the Paris Agreement. Among the most challenging sectors to decarbonize are the so-called hard-to-abate industries, which include steel, cement, chemicals (e.g., ammonia and methanol), and long-distance transportation. These sectors are characterized by their high energy intensity, reliance on fossil fuels, and process-related CO₂ emissions that are difficult to eliminate through conventional energy efficiency measures alone. Collectively, hard-to-abate sectors contribute to over 30% of global CO₂ emissions, making their transformation essential for achieving net-zero goals (Bataille 2020).

The steel industry, in particular, plays a pivotal role in global infrastructure development, yet it is one of the largest industrial contributors to greenhouse gas (GHG) emissions. In 2022, the steel sector was responsible for over 8% of global GHG emissions and more than 11% of global CO₂ emissions (IEA 2022). This is largely due to the carbon-intensive nature of traditional steelmaking processes and the sheer scale of global production, which more than doubled between 2000 and 2022, reflecting the growing demand for steel in construction, transportation, and manufacturing (Material Economics 2019). Steel is primarily produced through three technological routes:

1. Blast Furnace–Basic Oxygen Furnace (BF–BOF): This is the dominant method, accounting for over 70% of global steel production. It relies heavily on coal as both a fuel and a reducing agent, resulting in CO₂ emissions of approximately 2.3 tons per ton of steel produced (Hasanbeigi 2012).
2. Direct Reduced Iron–Electric Arc Furnace (DRI–EAF): This route contributes around 5% of global steel production and has a lower carbon footprint, with emissions of 1.4 tons of CO₂ per ton of steel (Vogl 2018).
3. Scrap-based Electric Arc Furnace (EAF): This method uses recycled steel and is the least carbon-intensive, producing 0.7 tons of CO₂ per ton of steel, and accounts for about 25% of global steel output (Worrell 2011).

From a decarbonization perspective, each of these pathways presents unique challenges and opportunities. A range of emerging technologies is under development or demonstration, including carbon capture and storage (CCS), electrification, and hydrogen-based reduction. Among these, the hydrogen-based DRI process stands out as one of the most promising solutions due to its potential to eliminate fossil fuel use in iron ore reduction, thereby drastically reducing emissions (Richardson-Barlow 2022).

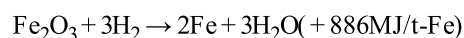
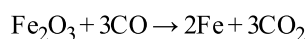
Table 1—List of low-carbon steel making processes (Witecka 2024).

Low carbon steel making process	Description	TRL	Expected readiness	Energy demand (GJ/t-crude steel)	CO2 reduction % wrt baseline	Cost of CO2 abatement-2030 (\$/tCO2)	Steel production cost (\$/t-cs)
H2-DRI-EAF	Hydrogen used as reducing agent in DRI unit, thus avoiding use of fossil fuels such as natural gas	8-9	2025-30 (100% H2)	10.8 (of which 8.3 GJ H2/t-cs)	-99.6%	137-192	727-857
NG-DRI-CCS	Use of carbon capture process for the flue gases coming from DRI and EAF	8-9	2025-30	12.2 (of which 10.5 GJ-NG/t-cs)	-89%	88-144	618-739
MOLTEN OXIDE ELECTROLYSIS (MOE)	Reduction of iron ore by electrochemical process	3-4	2035-40	12.4–14.8 (all electricity)	-100%	56-146 (by 2050)	582-766
BF-BOF-CCS	Use of carbon capture process for the flue gases coming from Blast Furnace and BOF	3-4	2030-35	22.8 (of which 19.5 GJ-coal/t-cs)	-73%	93-163	599-721
NZE-Scrap-EAF	Use of renewable electricity to power scrap EAF operation	9 (7 with charcoal)	Ready	2.8 (of which 2.5 GJ-coal/t-cs)	-99.6%	-	639-837

Recent studies and techno-economic analyses have highlighted the feasibility of hydrogen-based steelmaking, particularly in regions with abundant renewable energy resources. The integration of electrolyzers, hydrogen storage, Battery Energy Storage Systems (BESS), and photovoltaic (PV) plants into steel production facilities is being actively explored to enable green hydrogen generation and flexible energy management (Lechtenböhmer 2016). These systems not only support decarbonization but also enhance energy security and operational resilience.

The Direct Reduced Iron (DRI) production process

The Direct Reduced Iron (DRI) process is a key component of low-carbon steelmaking, offering a viable alternative to the traditional blast furnace route. In this process, iron ore is chemically reduced from its oxide form to metallic iron using a reducing gas, typically a mixture of carbon monoxide (CO) and hydrogen (H₂). The reduction takes place inside a shaft furnace, where the iron ore—usually in the form of pellets—is exposed to the reducing gases at temperatures below the melting point of iron. Traditionally, the reducing gas is produced by reforming natural gas, which generates CO and H₂. These gases react with iron oxides (primarily Fe₂O₃) to produce direct reduced iron, as per the following reactions:



A carburization step is also included in the process, where additional natural gas is injected to increase the carbon content of the final product, which is necessary for downstream processing and to meet metallurgical specifications. The DRI is then either cooled and stored or directly fed into the Electric Arc Furnace (EAF) in hot condition, which improves energy efficiency and productivity.

While the DRI–EAF pathway is often discussed as a whole, this paper focuses specifically on the DRI portion, which is the most energy-intensive and emissions-critical step. The EAF stage, which involves

melting the DRI, is not addressed in detail here. The decarbonization of the DRI process hinges on replacing natural gas with pure hydrogen as the reducing agent. This transition eliminates CO₂ emissions from the reduction reactions, as hydrogen reacts with iron ore to produce water vapor instead of carbon dioxide. According to the Intergovernmental Panel on Climate Change (IPCC), hydrogen is a key enabler for deep decarbonization in hard-to-abate sectors, including steelmaking (IPCC 2022). The hydrogen-based DRI process (H₂-DRI) can achieve over 99% reduction in CO₂ emissions compared to the conventional route (Vogl 2021).

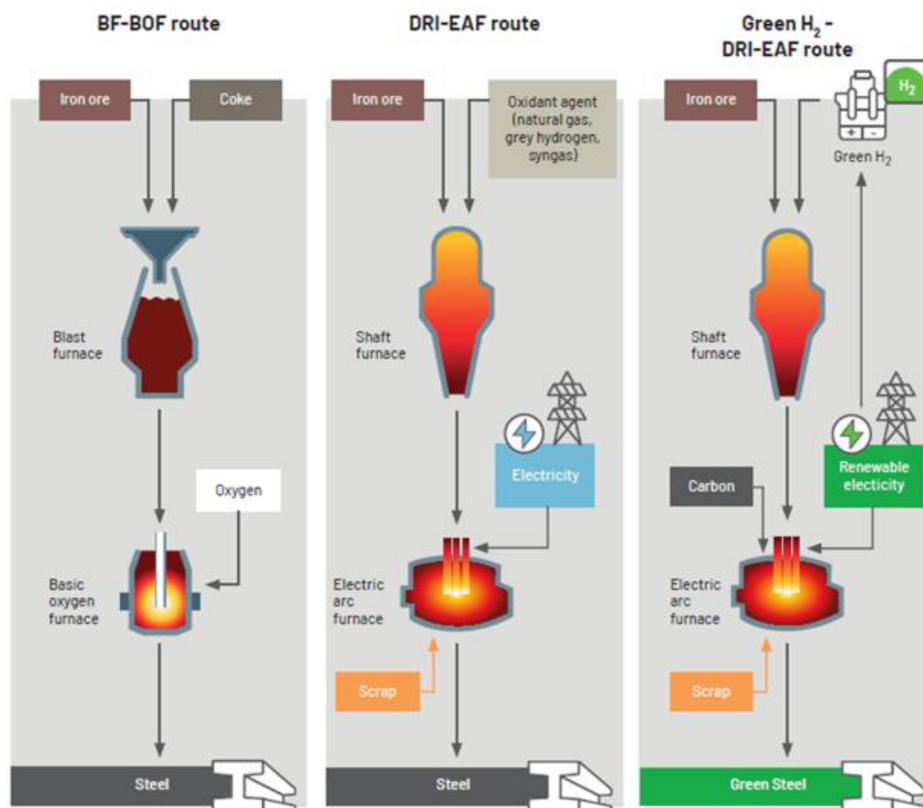


Figure 1—The production process for BF-BOF, natural gas DRI-EAF, and H₂-DRI-EAF (Hasanbeigi 2024).

The energy demand for hydrogen-based DRI is substantial, with estimates ranging from 2 to 4.5 MWh per ton of iron, depending on process efficiency and hydrogen purity. This underscores the importance of system-level optimization, including the sizing and operation of electrolyzers, storage systems, and renewable energy inputs. The use of Battery Energy Storage Systems (BESS) and PV plants can help manage intermittency and reduce reliance on grid electricity, making the process more sustainable and cost-effective. In conclusion, the hydrogen-based DRI process represents a transformative opportunity for the steel industry. By focusing on the DRI stage, this paper aims to optimize the design of integrated hydrogen production and storage systems to enable fossil-free ironmaking, paving the way for scalable and economically viable decarbonization.

Hybrid power plants: optimal design for Total Annual Cost (TAC) minimization

Traditional design heuristics, principles, and operational strategies developed for fully fossil-fueled energy systems are inadequate for hybrid configurations where gas turbine generators (GTG) are installed in parallel with Renewable Energy Sources (RES), BESS, electrolyzers, and hydrogen storage. This is due

to the increased number of components, the diversity of technologies involved, the complexity of their interactions, and the variety of potential system layouts (Zelaschi 2016). Consequently, optimizing the design and operation of hybrid power systems necessitates the use of advanced mathematical optimization techniques.

Although various commercial software solutions are available for hybrid system design optimization, they are typically focused on small, grid-connected microgrids that rely mainly on diesel generators. These tools often lack the capability to accurately model large, systems that utilize SCGT and CCGT potentially fed with variable fuels blending, and which must meet strict availability requirements. In partnership with a prominent research institution, BH has recently implemented an advanced optimization platform (Chong 2013) capable of supporting the optimal configuration of diverse hybrid systems. This platform can precisely simulate complex Combined Heat and Power (CHP) plants, apply constraints to ensure high availability and dependable operation, and model essential parameters based on environmental conditions (Zelaschi 2016).

The primary objective of the optimization is to determine the system configuration that minimizes the Total Annual Cost (TAC), while ensuring energy balance and compliance with all reliability and operational constraints, including fault tolerance. TAC encompasses both operational expenditures and capital investment costs, adjusted using the Capital Recovery Function (CRF), and is calculated as: $TAC = OPEX + CAPEX \cdot CRF$. This metric corresponds to the Levelized Cost of Energy (LCOE), given by $LCOE = TAC/YEC$ (Yearly Energy Consumption).

The optimization process requires the following inputs: i) thermal and electrical energy demand profiles; ii) renewable energy generation and ambient condition data; iii) performance and cost parameters for dispatchable units (SCGT/CCGT/Electrolyzer), battery energy storage systems, H₂ storage and photovoltaic arrays. The outputs include: i) the selection and sizing of dispatchable, non-dispatchable, and storage components; ii) the operational status and optimal scheduling of these units, including storage systems (e.g., on/off states, start-up/shutdown events, and hourly power set-points).

Hybrid power plant technologies for DRI steel facilities

This study focuses on the decarbonization potential of a Direct Reduced Iron steel manufacturing facility located in Milan, Italy. The plant has an annual production capacity of 2 million metric tons, equivalent to a continuous throughput of approximately 228.31 tons per hour. The facility's electrical energy demand is substantial, with a specific consumption of 110 kWh per ton of steel, resulting in a total constant power requirement of 25.114 MW. In addition to electricity, the plant is supposed to have thermal energy needs for both heating and cooling: heating demand peaks at 2 MW, primarily for space heating and process-related thermal load, while cooling demand reaches up to 1.5 MW, with dynamic daily profiles that vary seasonally and reflect realistic operational conditions, ensuring accurate representation of the plant's year-round energy behavior. Lastly, natural gas (and/or H₂) is needed as fuel and reducing agent for the DRI furnace: to remove the oxygen from the iron ore, in fact, a reducing gas is required, which can also be a mixture of natural gas and hydrogen. The higher the hydrogen percentage, the higher the decarbonization of the DRI production process. However, depending on the gas mixture composition, the overall equivalent energy requirement of the chemical process changes, as shown in, where the blue dots represent the experimental data taken from (Duarte 2024), while the specific CO₂ emissions in the case of 100% NG - 0% H₂ blend is taken from (Martinez 2025).

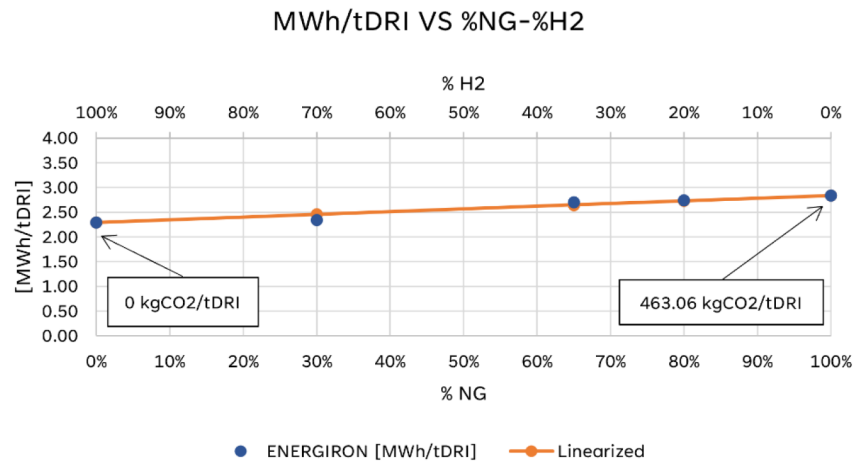


Figure 2—Equivalent overall energy requirement and carbon dioxide emissions of the CRI reduction process as a function of the natural gas/hydrogen blend.

This use-case was selected to provide a representative example of the challenges and opportunities in decarbonizing energy-intensive industrial operations. It allows for the exploration of hybrid energy systems, integration of renewable sources, and advanced control strategies to meet stringent reliability and efficiency requirements while reducing carbon emissions.

On the power generation side, to enhance the flexibility and responsiveness of the power island, gas turbines coupled with synchronous generators are considered among the top choices for supplying electricity to industrial facilities (Poullikkas 2005). This study specifically examines a configuration based on the NovaLT16, selected for its robust technical performance. The Baker Hughes NovaLT16 offers a simple cycle efficiency exceeding 37%, positioning it among the most efficient turbines in its class. Its fast ramp-up capability makes it particularly suitable for hybrid power systems that integrate renewable energy sources. Additional efficiency improvements beyond those of a Simple Cycle Gas Turbine (SCGT) can be achieved through a Combined Cycle Gas Turbine (CCGT) setup, where the turbine's exhaust gases are utilized to drive a steam turbine generator (Gambini 2023, Poullikkas 2005). The CCGT configuration adopted by Baker Hughes boosts plant flexibility and ensures optimized performance and emissions even during open cycle operation, which is common in various phases of a plant's lifecycle (Gambini 2023). For the analysis presented, the performance of the NovaLT16 under partial load conditions and varying ambient temperatures was simulated across SCGT, SCGT CHP, and CCGT CHP configurations using advanced commercial software tools. In each setup, the gas turbine generator (GTG) is modeled as a single-input, multiple-output system that consumes a hydrogen/natural gas blending (variable in time as per the needs of the plant, up to 100% hydrogen) and produces electricity, heat, and CO₂ emissions, which are function of the percentage of hydrogen consumed.

The PV plant under consideration is a single-axis tracking system, equipped with PV panels operating at 19% efficiency, inverters with 96% efficiency, and a DC/AC ratio of 1.2. Environmental conditions and the output of the renewable energy source (RES) were evaluated using the System Advisor Model (SAM) software (Blair 2014). A major limitation of the discussed optimization method is its reliance on hourly averages, which overlooks short-term variations like rapid intra-hour PV output fluctuations. These can significantly impact system reliability and control, with ramp rates up to 5 %/s and even exceeding 8 %/s reported in (Cormode 2013, De la Parra 2015, Marcos 2011, Sukumar 2018). To address this, some TSOs have imposed ramp rate limits on PV plants. A common mitigation strategy involves a dedicated Battery Energy Storage System (hereafter called BESS_{PV}) operating alongside the PV plant to smooth these fluctuations. BESS_{PV} sizing typically assumes a worst-case irradiance drop from 100 % to 10 % of the site's annual peak (De la Parra 2015): to avoid any impact on the main grid and on the GTG in operation, a possible

approach consists in relying solely on BESS_{PV} to manage PV variability. In this case, BESS_{PV} must be rated at 90 % of peak PV output power, with energy capacity sized for 30-minutes (conservative estimated time to startup a spare GTG, if needed). Considering end-of-life degradation (80 % capacity), a C2-rate BESS_{PV} rated at 113 % (0.9/0.8) of peak PV output per MWp is assumed. The added CAPEX is included with PV plant costs. Since most fluctuations are much smaller than the worst-case, the BESS_{PV} operates with small Depth of Discharge (DoD) cycles, supporting a 20-year lifespan without replacement.

The battery energy storage system available for the installation in the hybrid plant are based on a C-rate 1 (650 \$/MWh) and a C-rate 0.25 (500 \$/MWh) technology, both having an overall efficiency of 93% during charging and discharging operations.

To produce and store hydrogen at 200 bar, a system constituted by alkaline electrolyzer, H₂ compressor and hydrogen storage tank was considered, having investment costs equal to, respectively, 690 \$/kW (comprehensive of electrolyzer and compressor), and 16.62 \$/kWh. Efficiency maps of electrolyzer and compressor at partial loads were computed via commercial simulation software.

Operation and maintenance (O&M) expenses for all technologies evaluated in this study are based on actual current market values, including proprietary data from BH. For the financial modeling, assumptions included a 2% inflation rate, a 20-year expected plant life, and a nominal discount rate of 10%. We assumed to have 83 \$/tonCO₂ carbon tax, a natural gas price equal to 44 \$ and a variable electricity price taken from 2023 Italian electricity market (average 130 \$/MWh).

Finally, the Italian grid CO₂ intensity, 250 g/kWh, was considered to evaluate CO₂ emissions correlated with electrical energy purchased from the main grid.

Decarbonization analysis of a DRI production process

The design of an hydrogen production facility to satisfy the requirements of the described DRI plant could in principle be based on trivial considerations related to the constant flux of hydrogen needed to have the continuous DRI throughput of 228.31 tons per hour. For instance, assuming to analyze the case of complete decarbonization, 2.4 MWh/tonDRI are necessary for the reduction process. Adding the energy required for the process, heating and cooling, including an average 30% efficiency of conversion for Gas Turbine Generators and proper COP for the heat pumps, about 20 ton/h of hydrogen would be necessary, i.e. a constant flux of hydrogen having a power content of about 650 MW. Assuming an average electrolyzer efficiency of 50%, a 1.3 GW electrolyzer should be installed and run constantly at rated power. In terms of energy needed in one year, assuming as rule of thumb a capacity factor of 15 – 20% for a PV plant installed in the chosen location, a 7.5 GW PV plant would be necessary. Since the power output of renewable energy sources is extremely variable, however, the plant should be completed with a proper Battery Energy Storage System to flatten the PV fluctuations (considering both the day/night variations and the seasonal variations). This kind of solution, anyway, would be suboptimal, therefore by definition it would lead to a cost of hydrogen higher than the minimum one achievable.

To reduce the levelized cost of hydrogen, the optimal sizing of PV, BESS, electrolyzer and hydrogen storage has to be found, and this calls for the use of complex optimization algorithms able to consider the operation of the plant during each hour of a typical year. Figure 3 shows an overview of input-output models used in the DRI process optimization: the interconnections among the technologies presented in the previous paragraph are clearly visible.

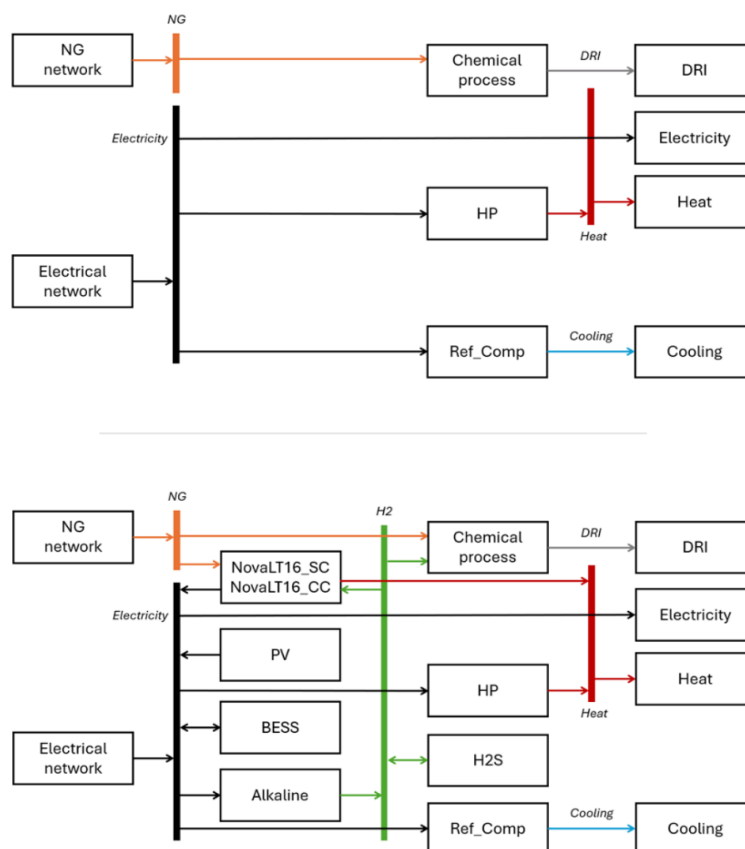


Figure 3—Input-output models overview used for DRI process optimization: baseline (above) and hybrid power plant designed to achieve increasing decarbonization targets (below)

The following table provides the main outcomes of several optimization problems, where a constraint imposing an increasing level of decarbonization was inserted in the model. It is possible to see all the main components of the decarbonizing infrastructure: i) gas turbine generators in simple cycle (SC) operated at 100% H₂, used to feed the facility, ii) size of PV plant; iii) size of alkaline electrolyzer; iv) size of 4h BESS (1h BESS was never installed); v) size of the hydrogen storage; vi) mix of natural gas (NG) and hydrogen in the reduction gas; vii) total carbon dioxide emissions; viii) total annual cost (TAC) of the infrastructure in per unit with respect to a benchmark case ("baseline") where all the required electrical energy is taken from the main grid and the reduced iron is produced using only NG.

Table 2—Main optimization results for several decarbonization targets

	Baseline –	20% decarb	40% decarb	60% decarb	80% decarb	100% decarb
GTG NovaLT16 [n°]	0	2	2	2	2	1
PV [GWp]	0	1.77	3.86	5.96	8.06	10.189
Alkaline [GWel]	0	1.06	2.35	3.65	4.94	6.15
BESS 4h [GWh]	0	0	0	0	0	0.40
H ₂ Storage [GWh]	0	41.31	89.44	137.61	185.79	233.70
NG – H ₂ mix [% NG]	100%	85,91%	66,81%	45,96%	23,10%	0%
NG – H ₂ mix [% H ₂]	0%	14,09%	33,19%	54,04%	76,90%	100%
Total CO ₂ emissions [ktonCO ₂ /y]	981,54	785,23	588,93	392,62	196	0
TAC (p.u.)	1	1,67	2,54	3,43	4,29	5,22

The dataset outlines a progressive transformation of a DRI (Direct Reduced Iron) production system, originally powered entirely by natural gas, toward full decarbonization through the integration of renewable energy and green hydrogen. This transition is not abrupt but unfolds across a continuum of increasing decarbonization levels, each requiring a corresponding scale-up in infrastructure and investment. At the outset, the system operates conventionally, with no renewable energy input. There are no photovoltaic installations, no electrolyzers, and no hydrogen storage. The reduction gas used in the DRI process is composed entirely of natural gas, and the total annual cost of energy infrastructure expressed in p.u. is by definition equal to 1. This baseline configuration results in nearly 1 million tons of CO₂ emissions per year, reflecting the carbon-intensive nature of fossil-based iron production.

As the system begins to decarbonize, the first noticeable shift is the introduction of PV capacity and alkaline electrolyzers, see Figure 4. These components are essential for generating and utilizing green hydrogen, which gradually replaces natural gas in the reduction process. At moderate levels of decarbonization, the system requires the installation of over 1.7 GWp of PV and more than 1 GWel of electrolyzer capacity.

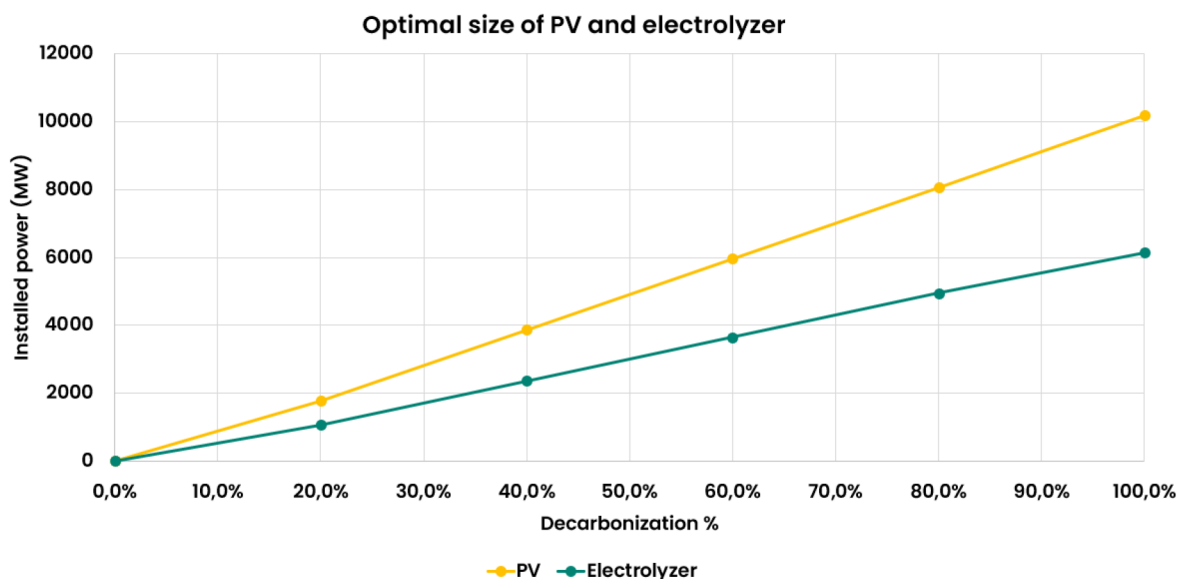


Figure 4—Optimal size of PV and electrolyzer to decarbonize the DRI process

To accommodate the intermittent nature of solar power and ensure a stable hydrogen supply, large-scale hydrogen storage is introduced, with capacities exceeding 40 GWh (Figure 5). This infrastructure enables the system to substitute around 14% of its natural gas consumption with hydrogen, reducing CO₂ emissions by 20%. However, this comes at a cost: the TAC increases by nearly 70%.

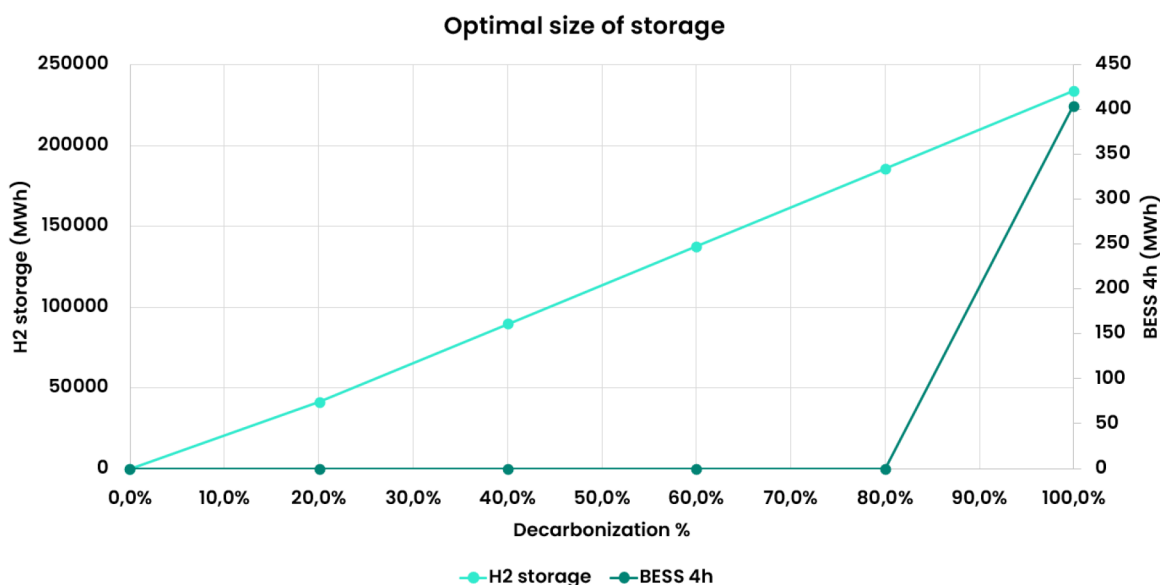


Figure 5—Optimal size of hydrogen storage and 4h BESS to decarbonize the DRI process

As decarbonization deepens, the infrastructure requirements grow substantially. The PV capacity more than doubles, and the electrolyzer capacity follows suit, reaching over 2.3 GW_{el}. Hydrogen storage also expands dramatically, now exceeding 89 GWh. These additions allow hydrogen to account for a third of the reduction gas, cutting emissions by 40%. The TAC, however, continues to rise, now approaching more than two and a half times the baseline.

Beyond this point, the system enters a phase where hydrogen becomes the dominant energy carrier. At 60% decarbonization, the PV capacity surpasses 5.9 GW_p, and electrolyzer capacity climbs to nearly 3.7 GW_{el}. Hydrogen storage grows to over 137 GWh. The fuel mix shifts decisively, with hydrogen now making up more than half of the reduction gas. This results in a 60% reduction in CO₂ emissions, but the TAC escalates to over three times the baseline.

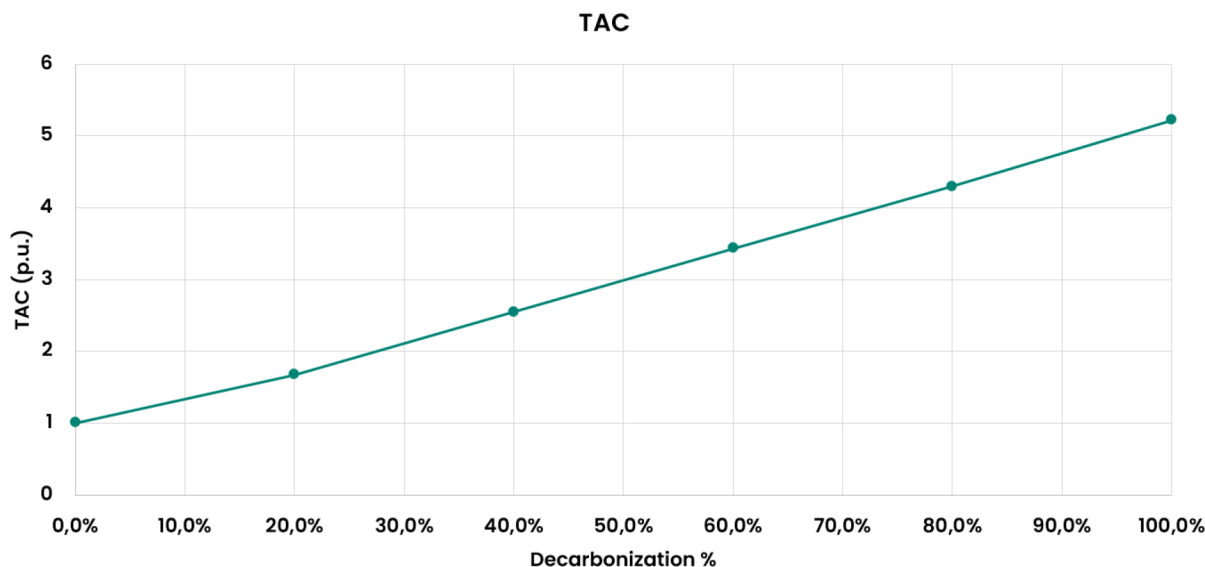


Figure 6—Total Annual Cost of the infrastructure needed to decarbonize the DRI process.

The final stages of decarbonization are the most demanding in terms of infrastructure. At 80% decarbonization, the system requires more than 8 GW_p of PV and nearly 5 GW_{el} of electrolyzer capacity.

Hydrogen storage must accommodate over 185 GWh. These massive installations enable hydrogen to replace over three-quarters of the natural gas, slashing emissions by 80%. The TAC, however, reaches \$1.59 billion, more than four times the original cost.

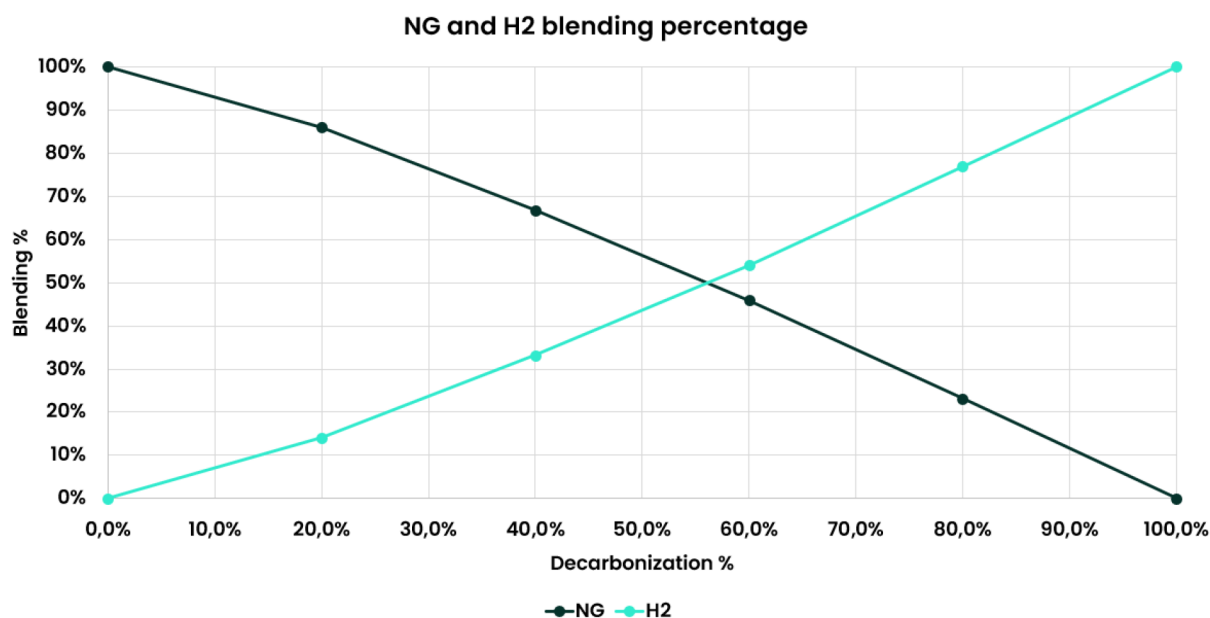


Figure 7—Natural gas and hydrogen blending (in percentage) for the reduction process.

Full decarbonization represents the culmination of this transformation. The system now operates entirely on hydrogen, with natural gas completely phased out. To support this, PV capacity must exceed 10 GW_p, and electrolyzer capacity must reach over 6 GW_{el}. Hydrogen storage peaks at nearly 234 GWh. The TAC climbs to over five times the baseline, but the reward is complete elimination of CO₂ emissions from the process. It is possible to see that the optimal solution is completely different with respect to the trivial one described at the beginning of the paragraph. The optimizer, in fact, basically avoids the installation of a BESS (only a small 0.4 GWth BESS is needed), minimizing the cost of hydrogen with an intermittent use of an "oversized" electrolyzer that is switched on only during the hours at which the output power of the PV plant is available. The BESS, when subject to charge/discharge cycling, is affected by wear, and needs to be replaced frequently (approximately, every 2500 – 3000 full discharge cycles considering Li-Ion BESS): this constitutes a major OPEX that disappears if the electrochemical storage is not present. The day/night balance is guaranteed by the hydrogen storage, that is used as seasonal storage as well, to flatten the differences between summer and winter production (this is indeed the main factor that drives the size of H₂ storage).

It is important to notice that the provided analysis is only focused on the energy requirements of the process (including into the "energy" needs even the NG or H₂ used for the iron reduction). To understand the impact of decarbonization on the overall cost of DRI production, we integrate this data with a baseline cost model (Table 3).

Table 3—Baseline cost model for a DRI production process.

Cost Component	Estimated Share (\$/ton) minimum	Estimated Share (\$/ton) maximum	Estimated Share (\$/ton) average	Source
Raw Materials (Iron Ore)	90	120	105	(Cavaliere 2019)
Natural Gas (Energy)	60	90	75	(Cavaliere 2019, Cavaliere 2022)
CAPEX (Depreciation)	30	50	40	(Cavaliere 2022)
Operation & Maintenance	40	60	50	(Cavaliere 2019)]
Utilities (Electricity, Water)	10	20	15	(Cavaliere 2019)]
Disposal & Environmental	5	10	7,5	(Cavaliere 2019)]
Logistics & Handling	10	15	12,5	(Anderson 2002)
Contingency & Overhead	10	15	12,5	(Cavaliere 2022)

In the conventional setup, the cost of producing one ton of DRI is approximately \$317.50. This includes \$75 per ton for energy, with the remainder covering raw materials, capital depreciation, operations, utilities, and other overheads. As the system decarbonizes, the energy cost component increases in proportion to the TAC. At 20% decarbonization, the energy cost rises to \$125.46 per ton, pushing the total DRI cost to \$367.96. At 40%, the energy cost reaches \$190.76, and the total cost climbs to \$433.26. This trend continues, see [Table 4](#), with the cost per ton reaching \$499.83 at 60%, \$564.48 at 80%, and finally \$633.68 at full decarbonization. This nearly doubling of production cost reflects the capital-intensive nature of renewable hydrogen infrastructure. While the environmental benefits are substantial—culminating in zero emissions—the economic trade-offs are significant, even relying on optimally designed plants. The analysis underscores the importance of policy support, technological innovation, and economies of scale in making green steel production competitive.

Table 4—Adjusted DRI cost estimate as function fo decarbonization level.

Decarbonization Level	TAC (p.u.)	Adjusted DRI Cost (\$/ton)
0%	1.00	\$317.50
20%	1.67	\$367.96
40%	2.54	\$433.26
60%	3.43	\$499.83
80%	4.29	\$564.48
100%	5.22	\$633.68

Conclusions and recommendations

Decarbonizing hard-to-abate industries like steel production is a critical step toward achieving global climate goals. The steel sector alone accounts for roughly 7–9% of global CO₂ emissions, largely due to its reliance on fossil fuels in energy-intensive processes such as direct reduced iron (DRI) production. Transitioning from natural gas to green hydrogen in DRI offers a promising pathway to drastically reduce these emissions, but it comes with significant technical and economic challenges. The analysis of a six-step decarbonization pathway reveals that achieving full decarbonization requires a massive scale-up of renewable infrastructure. This includes installing over 10 GWp of photovoltaic capacity, more than 6 GW

of alkaline electrolyzers, and nearly 234 GWh of hydrogen storage. These investments enable a gradual shift in the reduction gas from 100% natural gas to 100% hydrogen, ultimately eliminating CO₂ emissions from the process. However, this transition comes at a cost: the total annual energy-related infrastructure cost (TAC) increases more than fivefold, even considering optimally sized technologies. When this increase is reflected in the cost of DRI production, the price per ton rises from \$317.50 to \$633.68—an increase of nearly 100%. The analysis underscores the importance of policy support, technological innovation, and economies of scale in making green steel production competitive.

Nomenclature

BESS	Battery Energy Storage System
BH	Baker Hughes
CAPEX	Capital Expenditures
CC	Combined Cycle
CCGT	Combined Cycle Gas Turbine
CHP	Combined Heat and Power
CRF	Capital Recovery Factor
DRI	Direct Reduced Iron
EMS	Energy Management System
GHG	Greenhouse Gas
GT	Gas Turbine
GTG	Gas Turbine Generator
LCCA	Levelized Cost of Carbon Abatement
LCOE	Levelized Cost Of Energy
LCOE	Levelized Cost Of Energy
MILP	Mixed-Integer Linear Programming
O&M	Operation & Maintenance
OPEX	Operational Expenditure
PMS	Power Management System
p.u.	per unit
PV	Photovoltaic
RES	Renewable Energy Sources
SAM	System Advisor Model
SC	Simple Cycle
SCGT	Simple Cycle Gas Turbine
SOC	State Of Charge
ST	Steam Turbine
TAC	Total Annual Cost
TSO	Transmission System Operator
YEC	Yearly Energy Consumption

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